

Core Principles of the Scientific Variables Ontology

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1. INTRODUCTION

Variables underpin everything we do in scientific research. We measure their values and store them in data sets. They appear in science equations that encapsulate our current state of knowledge and show how different variables are related to one another. (The Ideal Gas Law, $P V = n R T$, provides a simple but illustrative example of such an equation.) All computational models are driven by values of input variables and produce values for output variables, thereby giving us predictive power. Scientific variables are based on the foundation of observation and quantification (i.e., measurement, metrology). At a minimum, a variable is not well-defined until we know: (1) what "thing" was observed, and (2) what property of that thing is being quantified. But variables can also involve many other concepts like physical processes, units, mathematical operations, and many more.

Since values of variables are also the exchange items in scientific workflows that are passed between different digital resources (e.g. numerical models, data sets, and web services), they are central to the problem of interoperability (Peckham et al., 2013). Recall that the “I” in the acronym FAIR (see Wilkinson et al., 2016) stands for interoperability, but tends to receive less attention than the other three letters. To tackle the interoperability problem in a general manner, computer science tells us that a hub-and-spoke approach is the most practical and scalable option. This requires the introduction of a standardized and very expressive representation of scientific variable concepts (the hub) that is: cross-domain, rules-based, human readable, machine actionable, unique, and able to disambiguate closely related concepts. The Scientific Variables Ontology (SVO, 2019), which began with the CSDMS Standard Names (Peckham, 2014; CSN, 2015), was created to fill this need. It is the result of carefully analyzing how variables are used in a wide variety of different scientific models and data sets, distilling a set of general, core concepts and principles and then formalizing these as an ontology using Semantic Web best practices & tools. See Stoica and Peckham (2018, 2019a, 2019b).

2. CORE CONCEPTS: TERMINOLOGY

In order to capture a complete, unambiguous description of a scientific variable, we require a collection of core terms as well as a number of classes that provide additional context for concepts. We first need to identify what is being observed, which we will refer to as an **object** or a **phenomenon**. A phenomenon may be thought of as “something that can occur”, and an **event** can then be viewed as an instance of a phenomenon that occurs within a specific range of time and space, often given a name. For example, a hurricane is a phenomenon, and Hurricane Maria was an event. To fully prescribe the object/phenomenon, adjectives are often necessary, as ‘dissolved’ and ‘organic’ in “dissolved organic carbon”. It is also often necessary to provide additional **context**, such as containment within a medium (e.g., air, water, soil) or a larger, parent object or comparison to a theoretical or reference concept. SVO has a general set of classes for providing this context. The next essential aspect of defining a variable is the **property** that is being observed or measured. Properties can be quantitative (**quantities**), or qualitative (often quantified with an **index**). They can also describe either the state of the object/phenomenon or an aspect of some **process** that it is involved in. Again, adjectives are often necessary for complete disambiguation. In thermodynamics, for example, there are several distinct concepts of “heat capacity” (or thermal capacity) which can be distinguished with the adjectives: mass-specific, mole-specific, volume-specific, isobaric, isochoric, and isothermal. In science, it is also common to apply a mathematical **operation** to a quantity to create a new quantity (see Section 5). Describing a scientific variable may require using all of these concepts. The number associated with a variable (using some method of measurement or estimation

and specific units) is referred to as its **value**. Notice, however, that our goal is only to fully describe the variable concept. When this concept is utilized in some digital resource, there will also be measurement units, a method of measurement, and typically some type of discretization in space and/or time. SVO does not attempt to capture this related information, which may differ from one resource to another. Instead, serialized SVO names can be used as arguments to functions that return them for a given resource. Sometimes it is clear that a variable is associated with a specific value, but there is no way to measure that value directly. Examples include the mass of Earth’s moon and the number of carbon atoms in a coffee cup.

3. PROPERTY NAMES

The properties of objects and phenomena can be divided into two categories: quantitative properties --- also called **quantities** --- and qualitative properties. While both are observable, quantitative properties are those that can be quantified or compared directly with numbers, and they often have measurement units rooted in physics. Qualitative properties are not directly measurable using numbers and a physics-based system of units, although an index or scale can be introduced to assign numbers to them for the purpose of monitoring or comparison.

Quantitative Properties. The International System of Quantities (ISQ, 2022), or ISO 80000, is a companion standard to the International System of Units (also known as “SI units”). It provides a general, unifying system, rooted in physics, that can be used to assign numerical values and units to a very broad class of quantitative properties. This system identifies seven fundamental **base quantities** (see Table 1), none of which can be expressed in terms of the others, which form the basis of a vast system of **derived quantities**. These derived quantities span virtually every quantity known to physics. The dimension (i.e., “generic units”) of any derived quantity can be represented with the formula: $L^a M^b T^c I^d \Theta^e N^f J^g$, where the exponents a through g can be any real number (i.e., negative, positive or zero). Symbols with an exponent of zero may be omitted. The dimensions of some common derived quantities are: velocity [L T-1], force [L M T-2] (SI unit Newton), pressure [L-1 M T-2] (SI unit Pascal), energy [L2 M T-2] (SI unit Joule), power [L2 M T-3] (SI unit Watt), voltage [L2 M T-1 I-1] (SI unit Volt), and frequency [T-1] (SI unit Hertz). Note that **dimensionless quantities** are also possible, with dimension symbol 1. These often arise as the ratio of two quantities with the same dimension (e.g., slope = rise/run [L/L]), but include angles measured in radians and any integer **count**. The ISQ also recognizes some dimensionless quantities associated with logarithmic scales called **levels**, such as sound pressure level (decibel) and information entropy (nat).

Base quantity	Symbol	Dimension	SI base unit	SI unit
length	l	L	metre	m
mass	m	M	kilogram	kg
time	t	T	second	s
electric current	I	I	ampere	A
thermodynamic	T	Θ	kelvin	K
amount of substance	n	N	mole	mol
luminous intensity	Iv	J	candela	cd

Table 1. Seven base quantities of the International System of Quantities (ISQ). Sometimes “currency” is added as an eighth base quantity, which accommodates units like dollars per gallon.

Qualitative Properties, Indexes, and Indicators. Many of the properties that we associate with objects and phenomena cannot be directly quantified using a physics-based system like the ISQ. To clarify this, think of the properties you can associate with a person. Clearly, a person is characterized by many quantitative properties such as their mass, weight, height, femur length (biometrics), pupillary distance and income. But they also have many other properties which are not straightforward to quantify, such as their intelligence, attractiveness, gullibility, generosity, amiability, obesity, strength, fitness, and creditworthiness. In our own minds we may have some imprecise “scale” that we use to think about and compare such properties among people. But what

if we want to “measure” such properties in a more objective way for the purpose of monitoring or comparison? When some specific method is introduced to compute and assign a numerical value to such properties, it is usually called an index, scale, or score. There are countless examples. For intelligence, we may use some sort of test score, such as an SAT score or an Intelligence Quotient. For obesity, the Body Mass Index (BMI) is commonly used. For creditworthiness, most banks use the FICO Score. For economic health, a stock market index like the Dow Jones Industrial Average (DJIA) can be used. For remote sensing of vegetation, there is NDVI (Normalized Difference Vegetation Index.)

Indexes¹ (vs. indices) are typically computed from several other properties, and when used for this purpose those properties are often called **indicators**. There are called this because they are in some way *indicative* of the qualitative property we wish to quantify, but they are insufficient by themselves. The Body Mass Index already mentioned for measuring obesity, is computed from just two indicators, a person’s height and weight (and both are quantities). Clearly, obesity is not characterized by weight alone, because taller people tend to weigh more. BMI is computed as body mass [kg] divided by height squared using SI units [kg / m²]. The FICO Score is computed using 5 indicators: payment history (on-time payments), debt burden, length of credit history, types of credit used, and recent searches for credit. Their relative weights are: 35, 30, 15, 10, and 10%. The DJIA index uses the stock prices for 30 prominent companies as indicators of economic health or growth. It uses 30 because no single stock provides an adequate measure by itself. Many indexes have been introduced throughout the sciences, especially in socioeconomics and remote sensing. They serve as pseudo-quantities for monitoring and comparison and are applied to a wide variety of objects and phenomena, from people to markets to countries.

Property Name Rule. The vast majority of property names in English are generated by converting an adjective (a descriptive word) to a noun. In linguistics, the process of converting a word that is not a noun to a noun (e.g., verb, adjective, adverb) is called **nominalization**. This is typically achieved by modifying the last few letters of the word, often with “ness” or “ity” (e.g., attractiveness, generosity). An interesting set of examples from old English end in “th”: wide – width, deep – depth, long – length, strong – strength, wrong – wretchedness, high – height, broad – breadth, wealthy – wealth.

4. PROCESS NAMES

Objects and phenomena are often subject to (or participate in) some sort of process or action. For example, a glacier can melt, slide, sublimate, retreat, or calve. The things that a glacier can do, or that can happen to it, are described with verbs. This simple observation leads to the following, very robust rule for the words that can be used as process names.

Process Name Rule. Virtually all process names in English are generated by converting a verb (action word) to a noun (i.e., nominalization). This is almost always done by changing the ending of the word. The most common endings are shown in Table 2:

Process Quantities. We frequently wish to quantify some aspect of such a process, such as its rate, the temperature at which it occurs, its frequency, etc. This is very often done by pairing a process name with a simple property name to create a **process quantity**. This is an extremely common pattern. Here are just a few examples: dilution ratio, drainage area, escape speed, gestation period, identification number, inclination angle, inflation pressure (for tires), penetration depth, relaxation time, striking distance (martial arts), turning radius (car), and vibration frequency. For process names that end in “ing”, the ending is often dropped, as in: burn, creep, flow,

Ending	Examples
tion	absorption, convection, infiltration
sion	avulsion, conversion, dispersion
ing	melting, swimming, upwelling
age	drainage, seepage, storage
y	discovery, recovery
ance	acceptance, governance, maintenance
ment	alignment, improvement, recruitment

Table 2. Common process name endings.

¹ This is a valid plural of the word index. Indices are often associated with array notation.

lapse, melt, shear and tilt (e.g., melt rate, shear stress). Process quantities can be contrasted with **state quantities** that describe the state of an object/phenomenon, such as its mass, temperature, or any measure of its size.

5. THE IMPORTANCE OF OPERATIONS

Many of the quantities used by geoscientists are generated by applying some sort of mathematical or other operation to an existing quantity to create a new quantity. In English, we almost always insert the verbal delimiter “of” after these operations, which can be chained together. So serialized SVO variable names use the word “of” as a delimiter for operations. The CSN and SVO include a large collection of operations, such as those in Table 3.

Operation	Change to Units, U	Applies to Field Type	Returns Field Type
time_derivative_of	U T-1	any (component-wise)	same
area_integral_of	U L2	scalar	scalar
log_of	log(U)	scalar	scalar
x_component_of	U	vector	scalar
x_y_component_of	U	tensor	scalar
azimuth_angle_of	radians	vector	scalar
zenith_angle_of	radians	vector	scalar
divergence_of	U L-1	vector	scalar
gradient_of	U L-1	scalar	vector
curl_of	U L-1	vector	vector
laplacian_of	U L-2	scalar	scalar

Table 3. Common mathematical operations and their effect on units. Some operations only apply to a specific “field type” (i.e., scalar, vector, or tensor).

The SVO is the only system for describing variables that we are aware of that also includes most mathematical operations, recognizing that applying a mathematical operation to a quantity simply produces a new quantity with modified units.

6. FLUXES AND FLOW RATES

In physics, there are 7 main “root quantities” that are conserved, and these are used across the geosciences in models and data sets. These are: charge [C], energy [J], mass [kg], moles [mol], momentum [kg m s⁻¹], number [1], and volume [m³]. Let X be any of these 7 conserved quantity names, with units U, as indicated above. We then have the associated quantities shown in Table 4.

Quantity Name	Change to Units	Field Type
X_flux	U L-2 T-1	vector
X_flow_rate	U T-1	scalar
X_concentration	U L-3	scalar
X_fraction	U/U	scalar
X_ratio	U/U	scalar
X_diffusivity	L2 T-1	scalar
divergence_of_X_flux	U L-3 T-1	scalar
z_integral_of_X_flux	U L-1 T-1	vector (“unit-width”)
gradient_of_X_concentration	U L-4	vector
z_integral_of_X_concentration	U L-2	scalar (“content”)

Table 4. Quantities associated with the seven main quantities that appear in conservation equations and their associated units. Note that a “flux” is a vector quantity that adds “per unit area per unit time” to base units, and the suffix “rate” adds “per unit time”. Moreover, a “flow rate” can be defined as the **area integral of the normal component of a flux**.

This table provides a logically consistent, easy to understand and almost self-defining variable naming schema for virtually all flow-related quantities that occur in science. However, the term that is actually used within a discipline may be quite different. For example, in the context of electricity,

the terms that are used are *electric current density* (vs. *charge flux*), *electric current* (vs. *charge flow rate*), and *volume charge density* (vs. *charge concentration*).

The imprecise concept of “rainfall rate” can be quantified as either a *volume flux* (volume of water that falls per unit area per unit time, e.g., mm/hr) or a *mass flux* (mass of water that falls per unit area per unit time, e.g., kg m⁻² s⁻¹). The latter method is preferred by meteorologists because unlike the former, it does not depend on the density of the precipitation, which can change dramatically with temperature (think: rain, snow, sleet, hail, graupel, etc.)

7. CONCLUSIONS

The purpose of this extended abstract has been to provide a high-level overview of some of the key concepts in the Scientific Variables Ontology (SVO). The ontology itself has been formalized using Semantic Web best practices and tools, meaning those sanctioned by the W3C (World Wide Web Consortium), the main international standards body for the World Wide Web. SVO uses the W3C standards RDF (Resource Description Framework), OWL (Web Ontology Language, a family of knowledge representation languages), and SKOS (Simple Knowledge Organization System, a “crosswalk standard” for connecting vocabularies). SVO is available online (SVO, 2022) and its variables will soon have permanent URLs at w3id.org. It can be queried using SPARQL (SPARQL Protocol and RDF Query Language). Ideas from SVO are also being incorporated into the I-ADOPT Framework Ontology, (I-ADOPT, 2021).

8. ACKNOWLEDGMENTS

This work was funded by grants from the National Science Foundation (NSF) EarthCube initiative and the DARPA World Modelers program.

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